Tutorial – Week 12

White-box Testing

```
binarySearch(elem searchKey, elem* array, int arraySize, bool &found, int &index)
{
    int bot, mid, top;                              (1)
    index = -1;
    bot = 0;
    top = arraySize – 1;
    mid = (top + bot)/2;

    if (array[mid] == searchKey)                    (2)
        found = true;                               (3)
    else
        found = false;                              (4)

    while (bot <= top && !found)                    (5)
        {
        mid = (top + bot)/2;                        (6)
        if (array[mid] == searchKey)                (7)
            {
            found = true;                           (8)
            index = mid;
            }
        else if (array[mid] < searchKey)            (9)
            bot = mid + 1;                          (10)
        else
            top = mid – 1;                          (11)
     }                                              (12)
}                                                   (13)
```

The above is a binary search function written in C language. Answer the following questions base on the code.

Question 1

Draw a control flow graph corresponding to the preceding function.

Question 2

For the preceding function, derive a set of test cases that will cover branch testing.

Question 3

Design additional test cases that will cover multiple-condition testing.

Question 4

Determine the cyclomatic complexity of the preceding function.

Question 5

Derive a set of test cases that cover McCabe’s Basis Paths.